KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 204: THEORY OF STRUCTURES I

1^{ST} SEMESTER 2020/2021 TIME: 2 HOURS

INSTRUCTIONS:

- This paper contains five questions.
- Answer **Question ONE** and any other **TWO** questions.
- Only THREE questions, including number **ONE** should be answered.

QUESTION ONE (30 MARKS)

a) An overhanging beam of length 10 m rests on supports placed 7 m apart. The left hand end is overhanging by 2 m and the right hand end is overhanging by 1 m. The beam carries a uniformly distributed load of 8 kN/m over the entire length. It also carries two point loads of 12 kN and 16 kN at the right and left end of the beam respectively. Draw the SF and BM diagrams for the beam and find the points of contraflexure if any. (8 Marks)

b) Calculate support reactions for the structures shown in Fig. 1b(i) and 1b(ii). (9 Marks)

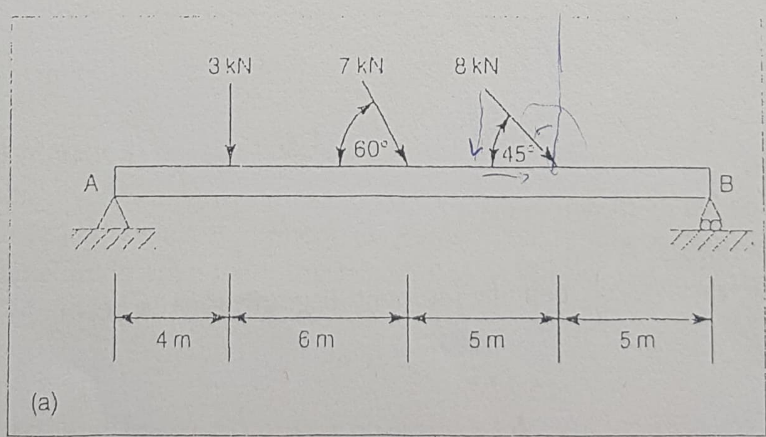

Fig. 1(i)

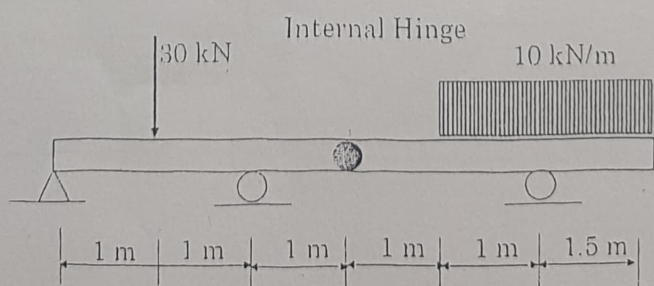

Fig. 1(ii)

c) For the truss shown in Fig 2, calculate the forces in all the members of the truss and indicate if each member is in tension or compression

(13 Marks)

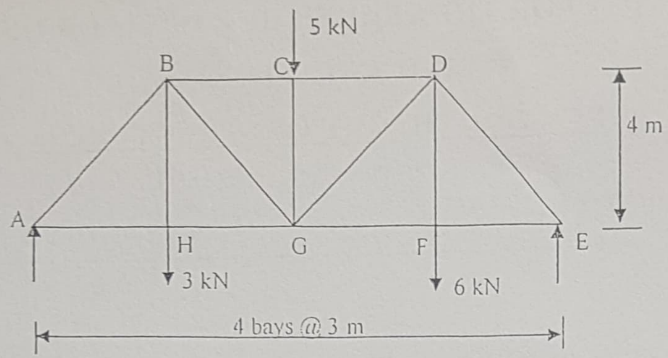

Fig. 2

QUESTION TWO (20 MARKS)

a) The simply supported beam in the accompanying illustration (Fig. 3) carries two concentrated load. Derive the expressions for the shear force and the bending moment for each segment of the beam in terms of x, the distance from support A, and use the equations to sketch the shear force and bending moment diagrams.

(10 Marks)

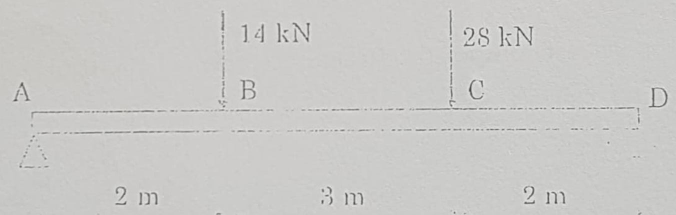

Fig. 3

b) A beam is loaded as shown in Fig 4.

i) Find reactions at A and B.

ii) Draw the shear force, bending moment diagrams for the beam.

(10 Marks)

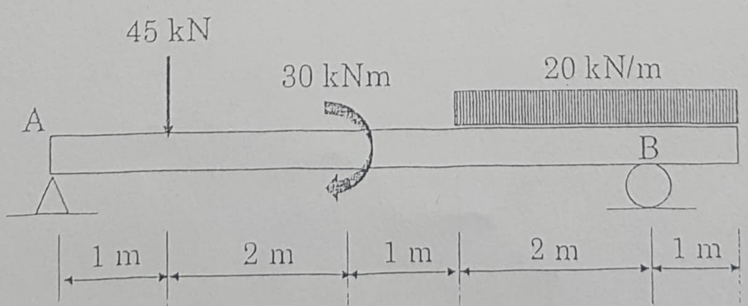

Fig. 4

QUESTION THREE (20 MARKS)

a) Determine the magnitude, sense and angle of inclination of the resultant of the following system of concurrent forces shown in Fig. 4.

(5 Marks)

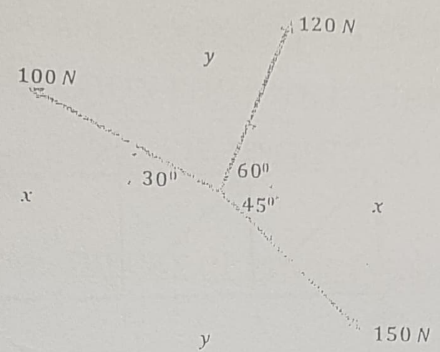

Fig. 4

b) Using the method of joint equilibrium, calculate the magnitude of the forces in all the members of frame ABCDEFG shown in Fig. 5 and state whether these members are in tension or compression. (15 Marks)

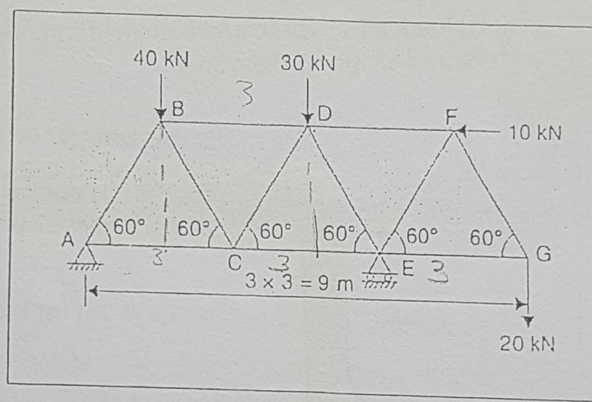

Fig. 5

225

QUESTION FOUR (20 MARKS)

a) Draw the shear force and bending moment diagrams for a simply supported beam of length L carrying a point load P at ¾ span from the left end of the beam and indicate maximum bending moment.

(5 Marks)

b) Draw the shear force and bending moment diagrams of the beam shown in Fig. 6 with internal hinges at C and E.

(15 Marks)

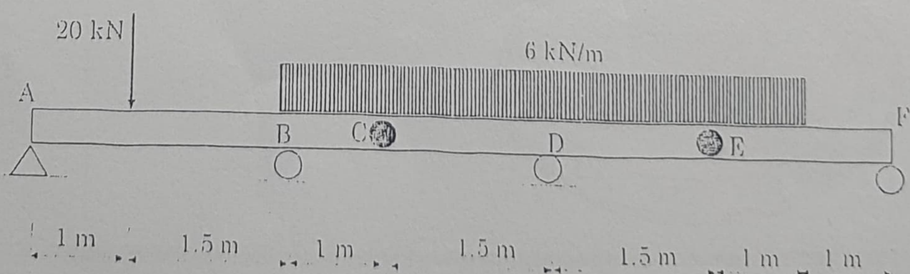

Fig. 6

QUESTION FIVE (20 MARKS)

a) A pin-jointed truss supported by a pinned support at A and a roller support at G carries three loads at joints C, D and E as shown in the Fig. 7. Determine the magnitude and nature (sense) of the forces induced in members X, Y and Z as indicated. (6 Marks)

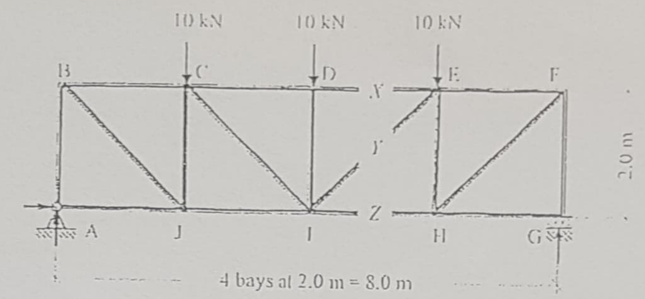

Fig. 7

b) A frame is supported on a roller and pinned as shown in the Fig. 8. For the loading indicated:

i) Determine the support reactions
ii) Sketch the axial load, shear force and bending moment
iii) Sketch the deflected shape of the frame

(14 Marks)

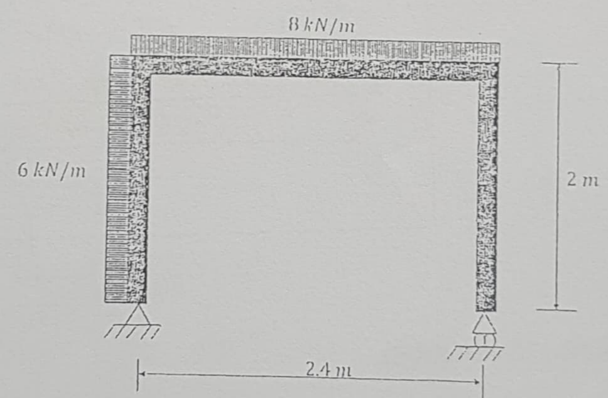

Fig. 8

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

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UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 204: THEORY OF STRUCTURES I

DATE: Tuesday, 19th March 2019 TIME: 8.00am - 10.00am

INSTRUCTIONS:

ANSWER QUESTION 1 AND ANY OTHER TWO QUESTIONS

Question One (30 Marks)

a) State 4 assumptions made in truss analysis. (2 Marks)

b) Derive an expression for maximum bending moment for a beam of span L, carrying a uniform load w over its entire span and simply supported at the ends. (4 Marks)

c) Determine the reactions for the figures shown in Fig 1(a) & 1(b).

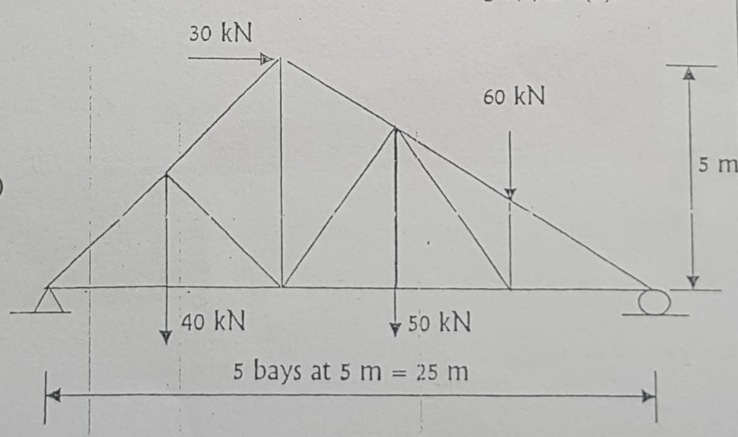
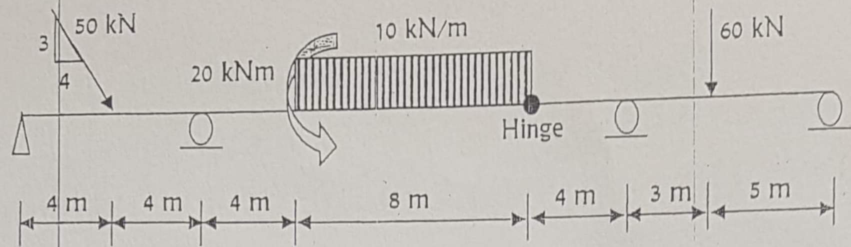


Fig. 1(a) (9 Marks)

Fig. 1(b)



(9 Marks)

(đ) Classify each of the following beams and/or frames (Fig. 2(a)-2(c)) as determinate or indeterminate. If statically indeterminate, determine the degree of indeterminacy. The members are subjected to external loadings that are assumed to be known and can act anywhere on the member.

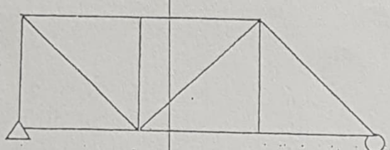


Fig. 2(a)

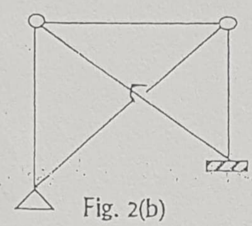


Fig. 2(b)

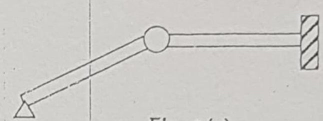


Fig. 2(c)

(6 Marks)

Question Two (20 Marks)

a) Draw a moment and load diagrams for the beam corresponding to the shear diagram shown in Fig. 3.

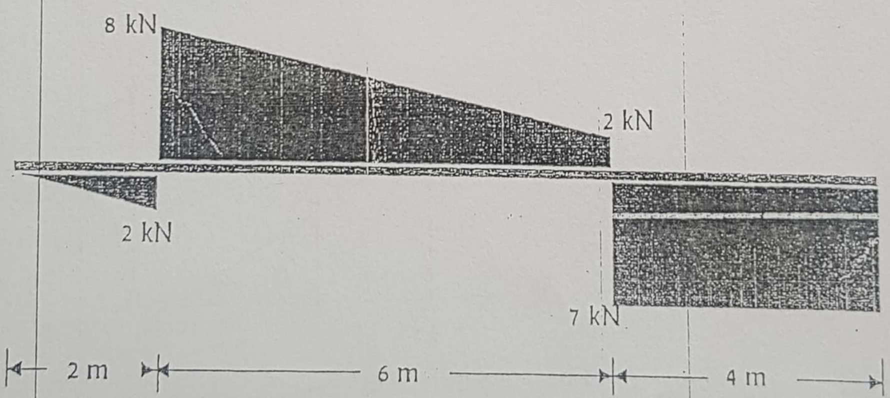


Fig. 3

(9 Marks)

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UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 204: THEORY OF STRUCTURES I

DATE: Monday, 26th February 2018 **TIME:** 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

Answer Q.1 and any other Two questions

Q.1 (a) Discuss the following terms used in structural analysis: (1 mark each)

(i) Ties → Members in tension.

(ii) Struts → Members in compression..

(iii) Beams and Girders → Beams are structures long as compared to their lateral dimensions

(iv) Columns → Vertical structures supporting compressive forces.

(v) Structural idealization — simplified

(vi) Truss → A structure composed of bars assumed to be connected by frictionless pins at their

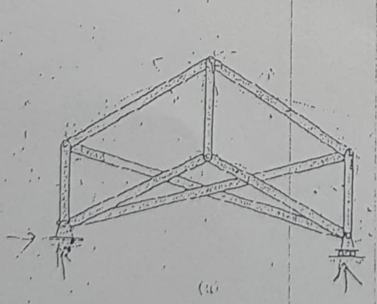
(vii) Line diagram - simplified diagram of the real structure for the purpose of analysis.

(viii) Live loads → loads that are moving or change their mass or weight.

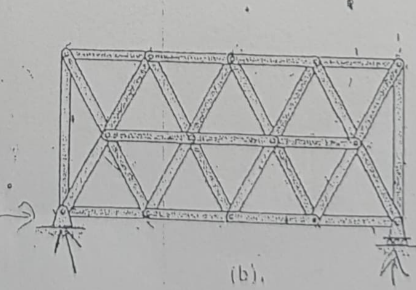
(ix) Dead loads → loads that are always there resulting from the weight of the structure

(x) Diaphragms

(b) Classify each of the following trusses as stable, unstable, statically determinate, or staticallyindeterminate. If indeterminate state degree of indeterminacy. All members are pin connected at their ends. (20 marks).

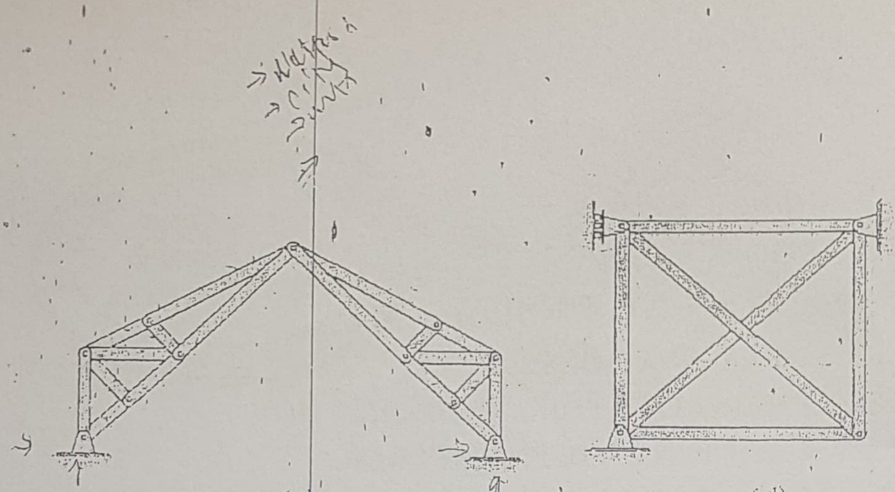


(a)



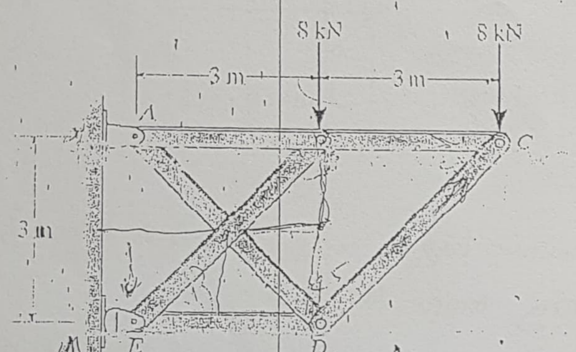
(b)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

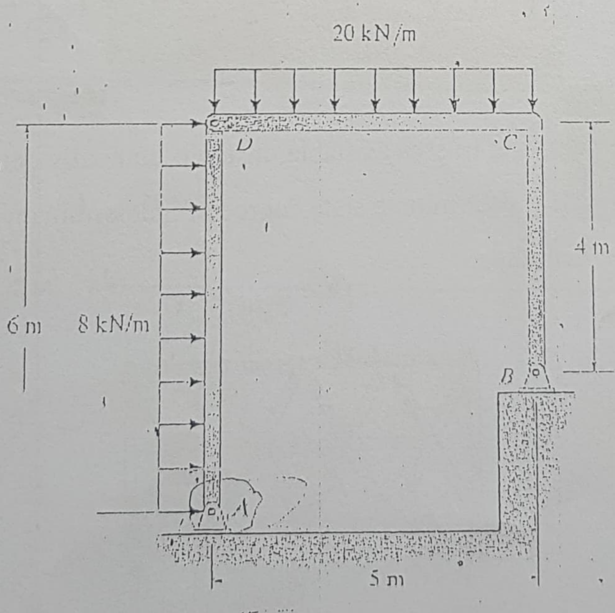


(c) (d)

Q.2. Determine the forces in each member ot the truss using method of joints. State if the members are in tension or compression. (20 marks)



Q.3. Determine the horizontal and vertical components of reactions at the supports A and B. (20 marks).



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (CIVIL ENGINEERING)

ECV 204: THEORY OF STRUCTURES I

DATE: Wednesday, 14th December 2016 **TIME:** 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

SECTION A (Compulsory)

Q.1 The side of the building in Fig.Q1. is subjected to a wind loading that creates a uniform normal pressure of 15kPa on the windward side and a suction pressure of 5kPa on the leeward side. Determine the horizontal and vertical components of reactions at A ,B and C of the supporting gable arch. **(30 marks)**

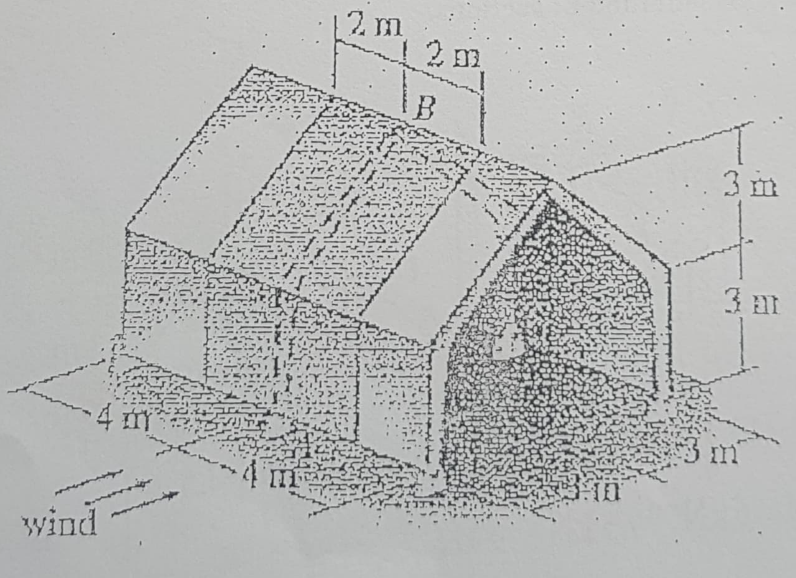


Fig.Q .1

= 2550
= 75kN
y = 300
x = 195kN

GY

SECTION B (Answer any two questions)

Q..2 Determine the force in each member of the scissors truss shown in Fig.Q2. State whether the members are in tension or compression. **Use methods of joints.** The reactions at the supports are given. (20 marks)

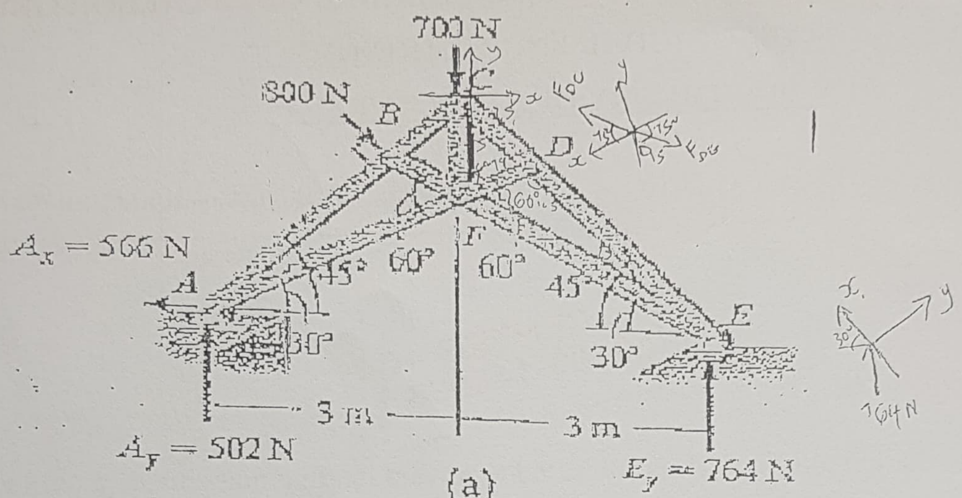


Fig.Q2

Q.3 Determine the forces in members BC, and MC of the K- Truss shown in Fig.Q3. . State whether the members are in tension or compression. The reactions are the supports have been calculated. **Use method of sections.** (20 marks)

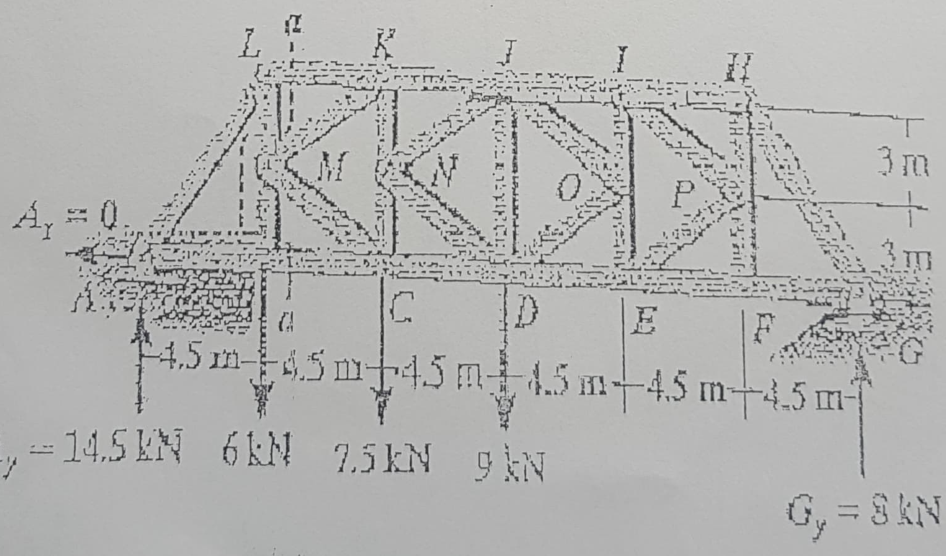
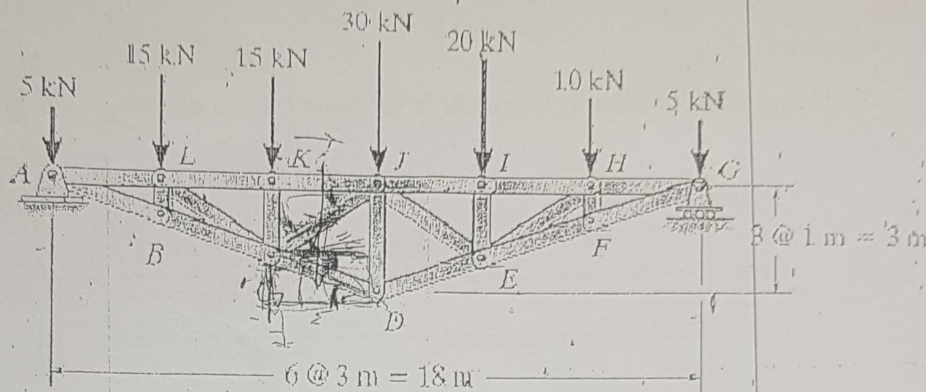


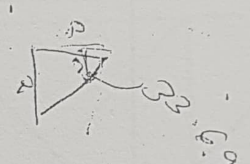
Fig.Q.3

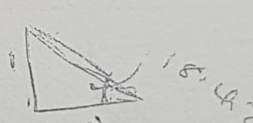
INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

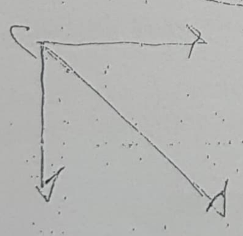
$F_{CF} = 2916N(C)$

Q. Determine the forces in members KJ, CD, and CJ of the truss using method of sections. State if the members are in tension or compression. (20 marks)









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DEPARTMENT OF CIVIL ENGINEERING

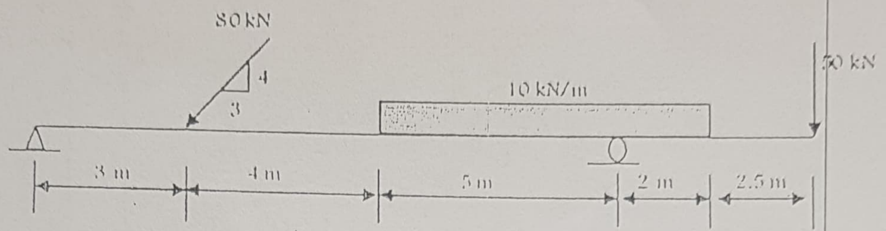
ECV 204: THEORY OF STRUCTURES I – CAT 1

TIME: 1HR 20 MINUTES ANSWER BOTH QUESTIONS

Question One

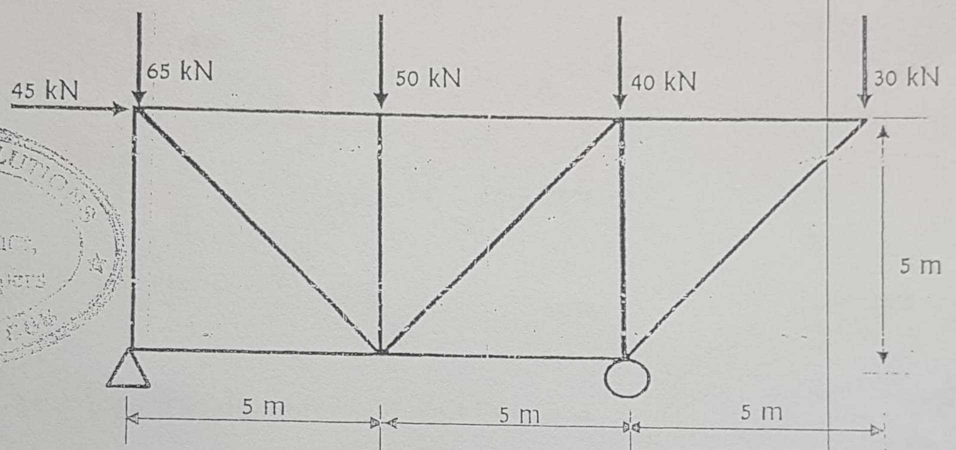
Find all reaction components in the structures shown below:

a)



(7 Marks)

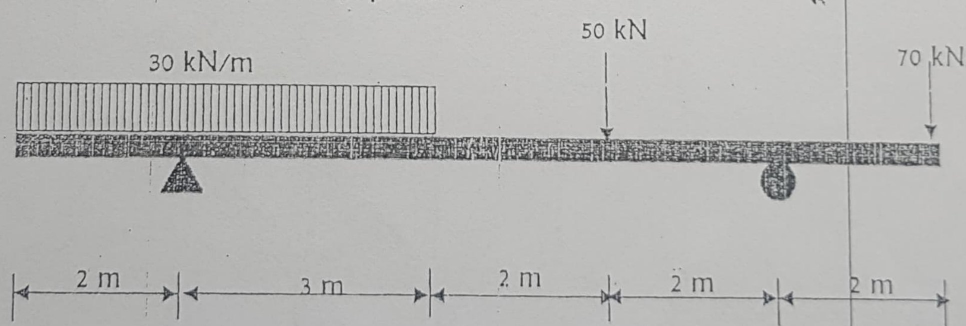
b)



(7 Marks)

Question Two

Sketch the shear force and bending moment diagrams for the beam loaded as shown in the figure below, indicating the values at critical points.



State the magnitudes and positions of maximum shear force and bending moment for this beam.

(16 Marks)

DEPARTMENT OF CIVIL ENGINEERING

CLASS OF 2017

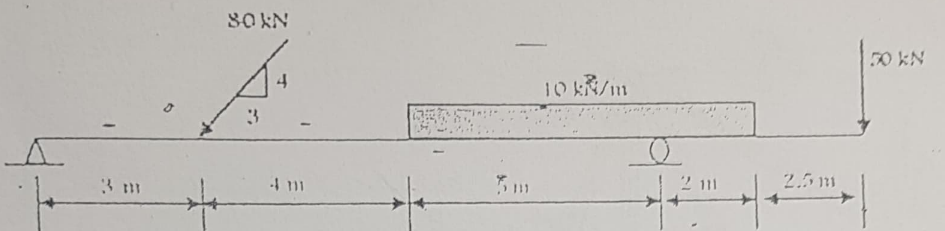
ECV 204: THEORY OF STRUCTURES I – CAT 1

TIME: 1HR 20 MINUTES **ANSWER BOTH QUESTIONS**

Question One

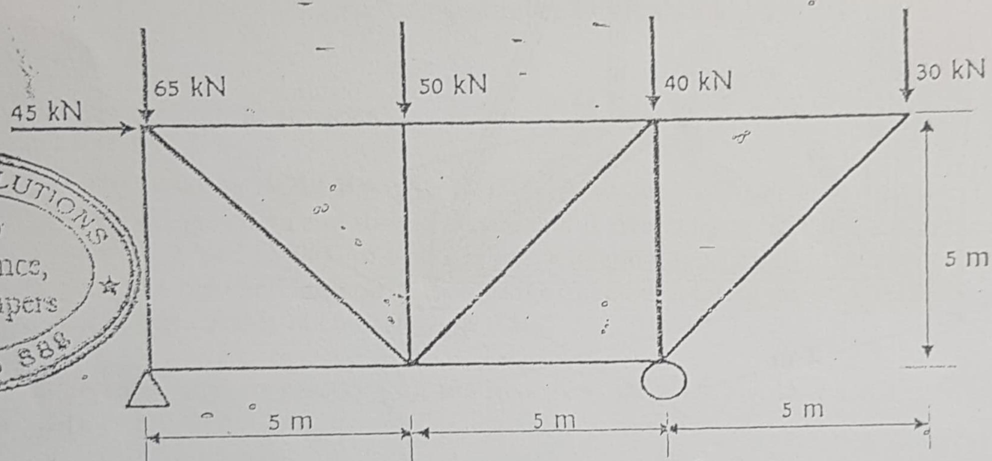
Find all reaction components in the structures shown below:

a)



(7 Marks)

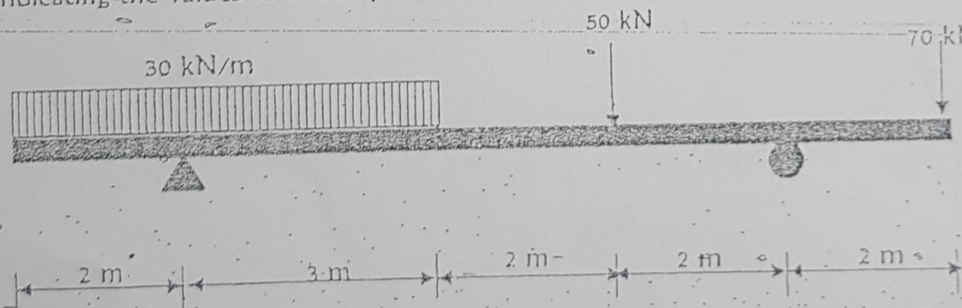
b)



(7 Marks)

Question Two

Sketch the shear force and bending moment diagrams for the beam loaded as shown in the figure below, indicating the values at critical points.



State the magnitudes and positions of maximum shear force and bending moment for this beam.

(16 Marks)

Q.4 Determine the force in each member of the space truss shown in Fig.Q.4. The truss is supported by a ball and socket joint at A ,a slotted roller at B , and a cable at C. Use **Tension coefficient method.** (20 marks)

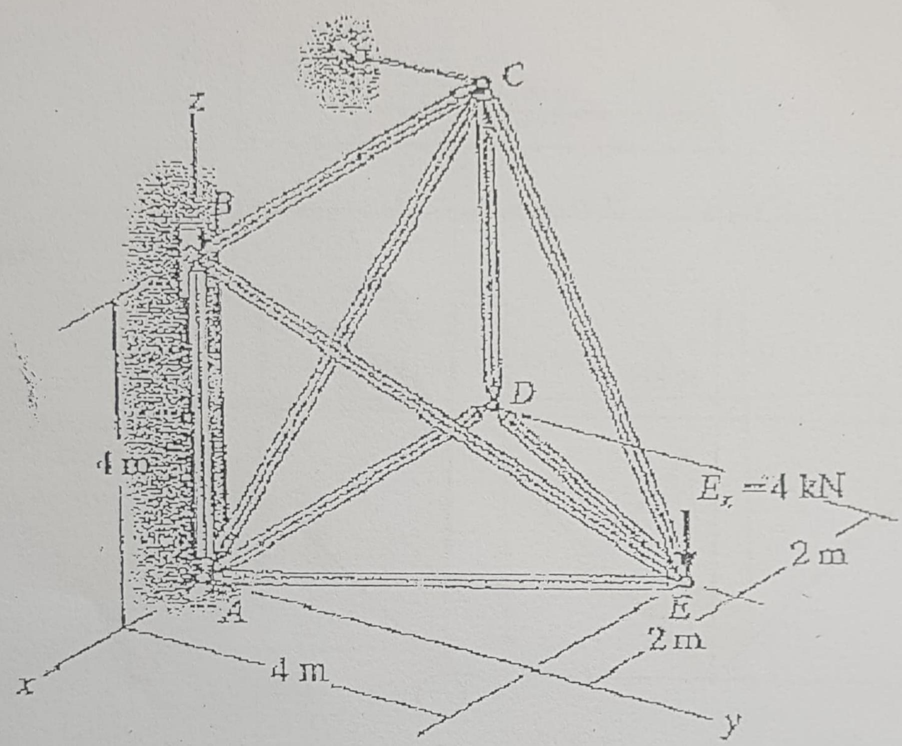


Fig. Q 4